Pauta R

a) a. R/ sabemos $A = \pi r^2 \longrightarrow A = 0.01\pi\ [m^2]$

Luego $\vec{A} = A\hat{u} \longrightarrow \vec{A} = \frac{0.01\pi}{\sqrt{2}}(\hat{\imath}+\hat{\jmath})\ [m^2]$

$$\vec{\mu} = I\vec{A} = (5)\left(\frac{0.01\pi}{\sqrt{2}}\right)(\hat{\imath}+\hat{\jmath}) \longrightarrow \boxed{\vec{\mu} = \frac{0.05\pi}{\sqrt{2}}(\hat{\imath}+\hat{\jmath})\ [A\cdot m^2]}$$ (3ptc

b. R/

$$\vec{\tau} = \vec{\mu}\times\vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{0.05\pi}{\sqrt{2}} & \frac{0.05\pi}{\sqrt{2}} & 0 \\ 0 & 5 & -5 \end{vmatrix}$$

$$= \left(-\frac{0.25\pi}{\sqrt{2}} - 0\right)\hat{\imath} - \left(-\frac{0.25\pi}{\sqrt{2}} - 0\right)\hat{\jmath} + \left(\frac{0.25\pi}{\sqrt{2}} - 0\right)\hat{k}$$

Luego: $\boxed{\vec{\tau} = \vec{\mu}\times\vec{B} = \frac{0.25\pi}{\sqrt{2}}(-\hat{\imath}+\hat{\jmath}+\hat{k})\ [N\cdot m]}$ (5ptos)

b) a. R/ segmento recto

$$\vec{F}_B = I_0\int d\vec{l}\times\vec{B}\ ; \quad d\vec{l} = dx\,\hat{\imath} - dy\,\hat{\jmath}, \quad \vec{B} = B_0\hat{k}$$

$$\vec{F}_B = I_0\int [dx\,(\hat{\imath}) - dy\,(\hat{\jmath})]\times[B_0\hat{k}] = I_0 B_0\left[-\int_{-L}^{0} dx\,(\hat{\jmath}) - \int_{-R}^{R} dy\,(\hat{\imath})\right]$$

$$\vec{F}_B = I_0 B_0\left[\left(x\Big|_0^{-L}\hat{\jmath}\right) - \left(y\Big|_{-R}^{R}(\hat{\imath})\right)\right] = I_0 B_0\left[-L\hat{\jmath} - (R-(-R))\,\hat{\imath}\right]$$

$\boxed{\vec{F}_B = -I_0 B_0(2R\hat{\imath} + L\hat{\jmath})\ [N]}$ (5ptos)

b. R/ segmento curvo

$$\vec{F}_B = I_0\int d\vec{l}\times\vec{B}\ ; \quad d\vec{l} = R\,\text{sen}\,\theta\,d\theta\,(-\hat{\imath}) + R\cos\theta\,d\theta\,(\hat{\jmath}), \quad \vec{B} = -B_0\hat{k}$$

$$\vec{F}_B = I_0\int [R\,\text{sen}\,\theta\,d\theta\,(-\hat{\imath}) + R\cos\theta\,d\theta\,(\hat{\jmath})]\times[-B_0\hat{k}]$$

$$\vec{F}_B = I_0 R B_0\int [\text{sen}\,\theta\,d\theta\,(\hat{\imath}) - \cos\theta\,d\theta\,(\hat{\jmath})]\times\hat{k}$$

$$\vec{F}_B = I_0 R B_0\left[-\int_{-\pi/2}^{\pi/3}\text{sen}\,\theta\,d\theta\,\hat{\jmath} - \int_{-\pi/2}^{\pi/3}\cos\theta\,d\theta\,\hat{\imath}\right] = I_0 R B_0\left[-\left(-\cos\theta\Big|_{-\pi/2}^{\pi/3}\right)\hat{\jmath}\right.$$

$$\left. - \left(\text{sen}\,\theta\Big|_{-\pi/2}^{\pi/3}\right)\hat{\imath}\right] = I_0 R B_0\left\{\left[\cancelto{1/2}{\cos\pi/3} - \cancelto{0}{\cos(-\pi/2)}\right]\hat{\jmath} - \left[\cancelto{\sqrt{3}/2}{\text{sen}\frac{\pi}{3}} - \cancelto{-1}{\text{sen}\left(-\frac{\pi}{2}\right)}\right]\hat{\imath}\right\}$$

$\boxed{\vec{F}_B = I_0 R B_0\left[-\left(1+\frac{\sqrt{3}}{2}\right)\hat{\imath} + \frac{1}{2}\hat{\jmath}\right]\ [N]}$ (5ptos)

C. R| $\vec{F}_{B\,Total} = \vec{F}_{B\,recto} + \vec{F}_{B\,curva}$

$$\vec{F}_{B\,Total} = I_0 B_0 \left[-\left(3 + \frac{\sqrt{3}}{2}\right) R\,\hat{\imath} + \left(\frac{1}{2}R - L\right)\hat{\jmath} \right] \; [N]$$ (2 ptos)